

Run 08 | OME

Global (including India)

IV + controls

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 100,087 (7.45%) **Case time** 53m 24s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9786). Highest F1: Random Forest (0.7144). For Random Forest, adding controls changed ROC-AUC by +0.0027 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9786	0.7464	95.05	62.65	83.11	0.7144	2m 50s
XGBoost	0.9782	0.7342	94.67	59.77	87.05	0.7088	24s
CatBoost	0.9778	0.7303	95.01	62.49	82.65	0.7117	3m 33s
H2O GBM	0.9763	0.7091	94.51	59.05	85.80	0.6995	1m 01s
Elastic Net	0.9744	0.6826	94.32	58.13	84.77	0.6897	14m 53s
Lasso	0.9744	0.6826	94.31	58.11	84.79	0.6896	13m 53s
Logistic Regression	0.9744	0.6820	94.32	58.19	84.79	0.6901	2m 08s
Ridge	0.9739	0.6747	94.34	58.25	84.90	0.6909	9m 48s
Decision Tree	0.9714	0.6443	94.35	59.09	78.82	0.6754	10s
Naive Bayes	0.9097	0.3835	89.46	38.89	72.38	0.5059	3m 54s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	30,600.38	35.32
2	Established business owner	11,840.54	13.67
3	Occupational status	4,781.42	5.52
4	Age	4,148.53	4.79
5	Size of government	2,082.49	2.40
6	GDP per capita (PPP)	2,002.08	2.31
7	Self-efficacy	1,916.52	2.21
8	Household size	1,873.83	2.16

Share is the percentage of total forest impurity importance across all 53 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Size of government	2,082.49	2.40
2	Legal system and property rights	1,773.48	2.05
3	Freedom to trade internationally	1,645.03	1.90

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 08_obj3_global_iv_cv_OME. Scores and importance are taken directly from saved CSV outputs.